POMS PRODUCTION AND OPERATIONS MANAGEMENT

PRODUCTION AND OPERATIONS MANAGEMENT

Vol. 30, No. 10, October 2021, pp. 3784–3799
ISSN 1059-1478 | EISSN 1937-5956 | 21 | 3010 | 3784
DOI 10.1111/poms.13481
© 2021 Production and Operations Management Society

Managing Perishable Inventory Systems with Age-differentiated Demand

Shouchang Chen
School of Management, Zhejiang University, Hangzhou, Zhejiang 310058, China, scchen@zju.edu.cn

Yanzhi Li
College of Business, City University of Hong Kong, Hong Kong, 852, China, yanzhili@cityu.edu.hk

Yi Yang, Weihua Zhou*
School of Management, Zhejiang University, Hangzhou, Zhejiang 310058, China, larryzhou@zju.edu.cn

We consider a periodic-review perishable inventory system with multiple demand classes, each characterized by a different lost-sales cost and the least freshness requirement. Demands of different classes in the same period could be correlated, while demands across periods are independent but not necessarily identical. In each period, the firm jointly makes the decisions regarding demand fulfillment, production/ordering, and disposal. The objective is to minimize the total discounted expected cost over the entire planning horizon including linear ordering cost, inventory holding/lost-sales cost, expiration cost, and disposal cost. By establishing new properties of multimodularity, we explore some monotonicity and bounded sensitivity properties of the optimal policies. The optimality analysis enables us to propose a novel approximation approach, called *adaptive approximation approach*, which can be recursively calculated through a single-dimension dynamic program. Numerical studies demonstrate that our proposed approximation approach is nearly optimal with the average optimality gap 0.30% and significantly outperforms the existing heuristics in the literature.

Key words: perishable inventory management; multimodularity; age-differentiated/age-dependent demand
History: Received: March 2020; Accepted: April 2021 by Annabelle Feng, after 1 revision.
*Corresponding author.

1. Introduction

Effective inventory management of perishable products is critical to business success and environmental sustainability. According to Buck and Minvielle (2013), fresh foods such as fruits and vegetables, meat, fish, dairy, and baked goods typically account for up to 40% of grocery chains' revenues, and they are also strong drivers of store traffic and customer loyalty. Meanwhile, one-third of all food produced for human consumption is wasted, with estimated associated costs equal to US$ 990 billion every year (FAO 2016). At the retailing level, for example, the percentage of waste in the United States is approximately 11% for dairy products and 4% for meat products (Buzby et al. 2014); for fruits and vegetables, waste is 8–9% in the United States and in Switzerland, while studies in Sweden, Norway, and Austria report waste in the range of 4–5% (Lebersorger and Schneider 2014). All these figures indicate that there is a significant room for improvement in managing perishable inventory.

Thanks to its importance, perishable inventory management has received considerable research attention for the past few decades. The existing research has largely focused on inventory systems with a single class of customers. However, customer behavior pertaining to perishable inventory is diverse and the market is often featured in multiple types of customers. As an attempt toward this direction, the present paper studies a perishable inventory system with age-differentiated demand.

The problem of our interest arises in different scenarios. First, it appears in business-to-business (B2B) contexts. In summarizing future research on perishable inventory management, Karaesmen et al. (2011) observed that in a B2B setting, "a service level agreement between a supplier and its retailer may not be as strict as 'freshest items must be supplied' (motivating last-in first-out, i.e., LIFO) or as loose as 'items of any age can be supplied' (motivating first-in first-out, i.e., FIFO), but rather 'items that will not expire within a specified time window must be supplied'." This leads to an inventory system with customers each requiring different least freshness for the delivered inventory. Blood platelets are often cited as an example of having multiple age-differentiated demand classes.

Treatment of patients for different diseases imposes a least requirement on platelet freshness. Moreover, patients requiring fresher platelets often have more critical conditions and thus are considered to be of higher priority. In this case, each class of demand is characterized by a least freshness requirement and a different shortage penalty cost. In the B2B context, Chen et al. (2019) consider a blood center which faces two classes of age-differentiated demand for platelets from downstream hospitals.

Second, online retailing, where the firm can control the inventory issuing sequence, is also featured in our problem. Many retailers now sell perishable products online. A prominent reason for consumer dissatisfaction with their shopping experience is that the delivered products are too close to, or even exceed, their expiration dates, as illustrated by the many negative comments on Amazon (http://www.amazon.com)[1] and JD (http://www.jd.com).[2] Due to reasons such as family size and shopping habit pertaining to package sizes, customers have different requirements for the least freshness of the received products. Notably, a customer's preference or expectation for freshness can be estimated from his/her purchasing history. Therefore, effectively, online retailers face multiple classes of customers differentiated by freshness requirement and possibly shortage penalty cost.[3]

Motivated by the two scenarios above, we study an inventory system that faces multiple classes of customer demand, each characterized by a least freshness requirement and a possibly different shortage penalty cost. Our model applies when the seller controls the inventory issuing sequence. When the different demand classes have the same shortage penalty cost, the inventory issuing sequence becomes irrelevant, and hence our model also applies even if customers control the inventory depletion sequence as in the ordinary grocery setting. In Online Appendix A, we present a stylized model for customers' different purchase behavior, suggesting FIFO, LIFO, or age-differentiated demand as different possibilities. In Online Appendix B, we show with a data set collected from two supermarket chain stores that our model provides a better reflection of the concerned customer behavior than the FIFO or LIFO model.

1.1. Our Contribution

In view of the wide applicability of the age-differentiated demand model for perishable products, in this study, we study the inventory control problem of a periodic-review perishable inventory system with multiple age-differentiated demand classes. Each demand class is associated with a potentially different lost-sales penalty cost. We assume that the firm can control the inventory issuing sequence. Our approach can also handle preset issuing policies such as FIFO and LIFO by imposing certain constraints on firm's issuing decision. In each period, the firm makes joint decisions regarding demand fulfillment, production/ordering, and disposal to minimize its total expected cost. Specifically, the paper makes the following contributions.

1. We characterize some monotonicity and bounded sensitivity properties of the optimal policies, as well as the properties of the objective functions. Along the way, we first establish some new preservation properties of multimodularity in the dynamic programming model, even when the feasible space is not a multimodular polyhedron, thus adding to the techniques of multimodularity.
2. We design a novel approximation approach, called *adaptive approximation approach*, which can be recursively calculated through a single-dimension dynamic program. The heuristic policy generated by this approximation approach retains the same structural properties as the optimal policy. Numerical studies demonstrate that our proposed approximation approach is nearly optimal with the average optimality gap 0.30% and significantly outperforms existing heuristics studied in the literature.
3. We are able to extend part of optimality analysis and apply our adaptive approximation approach to (single-class and the multiclass) LIFO perishable inventory system (see Section C in Online Appendix). The heuristic is also nearly optimal with the worst performance gap less than 0.51%.
4. Finally, when the shortage costs of different demand classes are identical, part of our optimality analysis and adaptive approximation approach can also be extended to the settings with sequential demand arrivals for both FIFO and LIFO systems (see Section C in Online Appendix).

1.2. Literature Review

In this subsection, we briefly review the related literature. The literature on perishable inventory management with the FIFO inventory issuing assumption is rich (Karaesmen et al. 2011, Nahmias 1982). The majority of this literature considers a single demand class, with Chen et al. (2014) being an exception. They study an inventory-pricing model with multiple types of demand, which are characterized by different shortage penalty costs, but not consider age-differentiated demand. The literature with the LIFO inventory issuing sequence is limited. Cohen and Pekelman (1978) are the first to study a LIFO

perishable inventory system. They characterize the age distribution of the steady-state inventory and study the critical number order policy. This study is followed by Pierskalla and Roach (1972), Deniz et al. (2010), Li et al. (2017), Fu et al. (2019) and Akkaş (2019).

For both FIFO and LIFO perishable inventory systems, due to the complexity of the optimal policy, a large amount of effort has been dedicated to developing effective heuristics, such as Nahmias (1976, 1977), Nansakumar and Morton (1993), Li et al. (2009), and Deniz et al. (2010). Recently, Chao et al. (2015), Zhang et al. (2016), Chao et al. (2017), and Zhang et al. (2020) have developed approximation algorithms for perishable inventory systems. In particular, Chao et al. (2015) point out that developing algorithms for perishable inventory with multiple types of age-differentiated demand is an important research topic.

There are a few studies that consider age-differentiated demand. Pierskalla and Roach (1972) are the first of such study and they investigate the optimality of FIFO issuing rule. Goh et al. (1993) model a system with two types of freshness demand and random supply and propose several methods to approximate this system. Ishii and Nose (1996) study a perishable inventory system with two types of customers: one depleting inventory in a FIFO manner and the other only accepts the newest inventory. Haijema et al. (2005, 2007) and Civelek et al. (2015) study the inventory management problem of blood platelets. Patients of different conditions require platelets of different freshness. The three works focus on the development of effective heuristic policies instead of analytical results. In particular, Haijema et al. (2005) and Haijema et al. (2007) allow demand for fresh inventory to be satisfied with old inventory by incurring a mismatching cost, that is, an upward substitution, whereas Civelek et al. (2015) allow both upward and downward substitution. None of the above works consider a general age-differentiated demand model. More importantly, we characterize the structure of the optimal policy and develop an effective heuristic based on the properties of the optimal policy.

Our work is closely related to Li and Yu (2014) and Abouee Mehrizi et al. (2019). In particular, we generalize the perishable inventory model by Li and Yu (2014) to multiple demand classes. Notably, existing properties regarding $L^\natural$-convexity/multimodularity cannot be directly applied to our problem. To overcome this challenge, we establish several new preservation properties of multimodularity. Abouee Mehrizi et al. (2019) study a very similar problem except that they assume a positive lead time and the unsatisfied orders are backordered instead of being lost. Moreover, Abouee Mehrizi et al. (2019) also made an attempt to solving the lost-sales case. However, because they consider the general case of positive lead time, multimodularity of the cost function cannot be established and thus they rely on numerical results to explore the structural properties of the optimal decisions. By contrast, we are able to characterize the structural properties by focusing on the case of zero lead time. Methodologically, we contribute by establishing some new preservation properties of multimodularity, even when the feasible space is not a multimodular polyhedron (see Lemmas 1 and 2), and they allow us to characterize the structure of the optimal policy succinctly. These new properties may be helpful for future research on other inventory systems. Most importantly, we design an efficient and easy-to-implement heuristic policy, called adaptive approximation, for both FIFO and LIFO issuing rules. This heuristic significantly outperforms the known heuristics for perishable inventory systems, including the one proposed by Abouee Mehrizi et al. (2019). Finally, we are the first to study LIFOM (the age-differentiated demand follows 'last-in first-out' rule) inventory system. We partially characterize the optimal policies and show that the adaptive approximation heuristic is also very effective for the LIFOM inventory system.

1.3. Structure

The remainder of this study is organized as follows. In section 2, we introduce the model and formulation. In section 3, we characterize the structure of the optimal policy by applying and extending multimodularity techniques. In section 4, we propose an adaptive approximation heuristic, which retains the structural properties of the optimal policy. Section 5 reports the numerical experiments, which illustrate the effectiveness of our heuristic and other existing heuristics. The paper concludes in section 6.

2. The Model and Formulation

In this section, we describe the model in detail. Throughout the paper, we define $\Delta_{x_i}^{\delta} f(\mathbf{x}) = f(\mathbf{x}+\delta e_i) - f(\mathbf{x})$ and $\Delta_{x_i} f(\mathbf{x}) = \partial f/\partial x_i$. We denote n-dimensional real space as $\mathbb{R}^n$ and its nonnegative counterpart as $\mathbb{R}_+^n$. The notation $\mathbf{e}_i \in \mathbb{R}^n$ represents the n-dimensional vector with the ith entry equal to 1 and other entries equal to 0. The binary operators $\vee$ and $\wedge$ are defined as $a \vee b = \max\{a, b\}$ and $a \wedge b = \min\{a, b\}$ for any $a, b \in \mathbb{R}$. For notational convenience, we use $[N]$ to

denote the set of {1, 2, …, N} for any positive integer N.

Consider a firm that manages a periodic-review single-product inventory system over a finite planning horizon of T periods. The product is perishable and has an n-period lifetime, that is, a new product will perish after n periods. In each period, unsatisfied demands are lost; this is a common assumption in the perishable inventory control literature (Fries 1975, Nansakumar and Morton 1993) and inventory rationing literature (Ha 1997). Suppose there are w different demand classes in the market, each characterized by a different lost-sale cost p^k and a least freshness requirement m^k. In other words, customers of class k can be satisfied only by products with remaining lifetimes no less than m^k periods. Without loss of generality, we assume that $p^1 \geq p^2 \geq \cdots \geq p^w$, that is, shortage costs are decreasing in the class number. However, we do not need to restrict the order of freshness requirements from different demand classes; the demand class with a higher shortage cost may have a higher or lower freshness requirement. Demand of class k in period t, denoted by D_t^k, is stochastic. Demands of different classes in the same period could be correlated, while demands across periods are independent but not necessarily identical.

The sequence of events in each period t is as follows. For notational convenience, we remove without confusion the time index of state and decision variables.

Stage 1: *(Demand Fulfillment Decision)* At the beginning of the period, all types of demand $\{D^k\}_{k=1}^w$ are realized. We assume that the firm can design the mechanism to prioritize the on-hand inventory to satisfy different types of demand. Define the initial inventory level in the current period as $\mathbf{x} = (x_1, x_2, \ldots, x_n)$, where x_i represents the on-hand inventory level with a remaining lifetime of i periods. After reviewing the inventory levels $\mathbf{x}$, the firm needs to determine the amount of inventory with a remaining lifetime of i periods to fulfill the demand class k, denoted by d_i^k. We define $\mathbf{d}^k = (d_1^k, d_2^k, \ldots, d_n^k)$ for $k \in [w]$ and call $\mathbf{d} = (\mathbf{d}^1, \mathbf{d}^2, \ldots, \mathbf{d}^w)$ the *demand fulfillment decision*. We next analyze the constraints on the demand fulfillment decision. First, recall that demand class k requires products with remaining lifetimes of no less than m^k. That is, the firm's fulfillment decision should satisfy $d_i^k = 0$ for any $i \in [m^k - 1]$. Second, the fulfillment amount should be nonnegative, that is, $d_i^k \geq 0$, for any $i \in [n]$ and $k \in [w]$. Third, the total fulfillment amount for demand class k must be no more than the realized demand, that is, $\sum_{i=1}^n d_i^k \leq D^k$ for any $k \in [w]$. Finally, the total consumption amount of inventory with i periods of remaining lifetime should be constrained by its on-hand inventory, that is, $\sum_{k=1}^w d_i^k \leq x_i$, for any $i \in [n]$. Overall, the feasible region of the demand fulfillment decision in period t can be expressed as

$$F_{\mathbf{d}}(\mathbf{x}, \mathbf{D}) = \left\{ \mathbf{d} \;\middle|\; \begin{array}{l} \sum_{i=1}^n d_i^k \leq D^k, \text{ for } k \in [w]; \\ \sum_{k=1}^w d_i^k \leq x_i, \text{ for } i \in [n]; \\ d_i^k \geq 0, \text{ for } i \in [n] \text{ and } k \in [w]; \\ d_i^k = 0, \text{ for } i \in [m^k - 1] \text{ and } k \in [w]. \end{array} \right\} \tag{1}$$

After the demand fulfillment decision, the unmet amount of demand class k is $D^k - \sum_{i=1}^n d_i^k$. Consequently, the total shortage cost incurred in the current period is $\sum_{k=1}^w p^k (D^k - \sum_{i=1}^n d_i^k)$.

Stage 2: *(Expiration)* After demand fulfillment, the leftover inventory level of the product with i-period lifetime is $x_i - \sum_{k=1}^w d_i^k$ for any $i \in [n]$. The products with a one-period remaining lifetime will perish at the end of the period, incurring an expiration cost at θ per unit. Therefore, the total outdating cost of expired products is $\theta(x_1 - \sum_{k=1}^w d_1^k)$. We define the on-hand inventory after expiration as $\mathbf{y} = (y_1, y_2, \ldots, y_{n-1})$, where y_i denotes the amount of on-hand inventory with a remaining lifetime of i+1 periods. Then, $\mathbf{y}$ can be expressed as follows:

$$y_i = x_{i+1} - \sum_{k=1}^w d_{i+1}^k, \text{ for any } i \in [n-1]. \tag{2}$$

Stage 3: *(Disposal Decision)* At the end of the period, the firm needs to make the disposal decision or, equivalently, decide how much inventory of different lifetimes to be carried over to the next period. Define the inventory level after disposal as $\mathbf{z} = (z_1, z_2, \ldots, z_{n-1})$, where z_i represents the inventory level of products with a remaining lifetime of $i + 1$ periods. The inventory level after disposal should be nonnegative and no more than the inventory level after expiration. That is, the decision $\mathbf{z}$ should be constrained by

$$F_{\mathbf{z}}(\mathbf{x}, \mathbf{d}) = \left\{ \mathbf{z} | 0 \leq z_i \leq y_i = x_{i+1} - \sum_{k=1}^w d_{i+1}^k, \text{ for } i \in [n-1] \right\}. \tag{3}$$

The products that are carried over to the next period incur a holding cost h per unit, and hence, the total holding costs can be expressed as $h\sum_{i=1}^{n-1} z_i$. For analysis convenience, we assume that the per-unit disposal cost is equal to the per unit outdating cost; the total disposal cost in period t is $\theta\sum_{i=1}^{n-1}(y_i - z_i)$. However, our results also hold when the disposal cost is no larger than the outdating cost. Note that allowing inventory disposal is a common assumption in the periodic-review perishable inventory literature; see Chen et al. (2014), Li and Yu (2014), Li et al. (2016), and Chen et al. (2019) for example. In particular, a negative disposal cost corresponds to the

clearance sales, a common practice in retailing, as considered by Li and Yu (2014) and Li et al. (2016). Moreover, allowing voluntary disposal brings technical convenience; as shown by Chen et al. (2014), without allowing intentional disposal, FIFO rule may not be optimal for a perishable inventory system.

Stage 4: *(Replenishment Decision)* Finally, after observing the post-disposal inventory level $\mathbf{z}$, the firm needs to make a replenishment decision with a per-unit ordering cost c. The order quantity should be nonnegative, that is, $q \in F_q = \{q|q \geq 0\}$. We assume that replenishment orders arrive instantaneously, that is, the replenishment lead time is zero.

The firm's objective is to minimize the total discounted expected cost over the entire planning horizon with the per-period discount factor $\alpha \in (0, 1]$. Given the initial inventory level $\mathbf{x}$, let $\widehat{V}_t(\mathbf{x})$ be the optimal discounted expected cost from period t until the end of the planning horizon. Then, the firm's optimization problem can be formulated as the following dynamic program:

$$\widehat{V}_t(\mathbf{x}) = \mathbf{E}\left\{\min_{\mathbf{d}\in F_{\mathbf{d}}(\mathbf{x},\mathbf{D}),\, \mathbf{z}\in F_{\mathbf{z}}(\mathbf{x},\mathbf{d}),\, q\in F_q}\left\{cq + g_t(\mathbf{x},\mathbf{D},\mathbf{d},\mathbf{z}) + \alpha\widehat{V}_{t+1}(\mathbf{z},q)\right\}\right\}, \tag{4}$$

where

$$g_t(\mathbf{x},\mathbf{D},\mathbf{d},\mathbf{z}) = \sum_{k=1}^{w} p^k\left(D^k - \sum_{i=1}^{n} d_i^k\right) + \theta\left(x_1 - \sum_{k=1}^{w} d_1^k\right) + \theta\sum_{i=2}^{n}\left(x_i - \sum_{k=1}^{w} d_i^k - z_{i-1}\right) + h\sum_{i=1}^{n-1} z_i,$$

and $\widehat{V}_{T+1}(\mathbf{x}) = 0$.

To facilitate the subsequent analysis, we reschedule the time periods. Define $V_t(\mathbf{y})$ as the optimal discounted expected cost beginning with the post-expiration state (i.e., after stage 2) in period t until the end of the planning horizon. In this way, the event sequence in the new period t is the disposal decision, the replenishment decision, the fulfillment decision, and expiration. With the new sequence of events, the problem can be reformulated as follows:

$$\begin{aligned}
V_t(\mathbf{y}) &= \theta\sum_{i=1}^{n-1} y_i + \min_{\mathbf{z}\in F_{\mathbf{z}}(\mathbf{y})} L_t(\mathbf{z}),\\
L_t(\mathbf{z}) &= (h-\theta)\sum_{i=1}^{n-1} z_i + \min_{q\in F_q} u_t(\mathbf{z},q),\\
u_t(\mathbf{z},q) &= cq + \alpha\mathbf{E}\left\{\min_{\mathbf{d}\in F_{\mathbf{d}}((\mathbf{z},q),\mathbf{D})} I_t((\mathbf{z},q),\mathbf{d})\right\},\\
I_t((\mathbf{z},q),\mathbf{d}) &= \theta\left(z_1 - \sum_{k=1}^{w} d_1^k\right) + \sum_{k=1}^{w} p^k\left(D^k - \sum_{i=1}^{n} d_i^k\right)\\
&\quad + V_{t+1}\left(z_2 - \sum_{k=1}^{w} d_2^k, \ldots, z_{n-1} - \sum_{k=1}^{w} d_{n-1}^k, q - \sum_{k=1}^{w} d_n^k\right),
\end{aligned} \tag{5}$$

where $F_{\mathbf{z}}(\mathbf{y}) = \{\mathbf{z}|\mathbf{0} \leq \mathbf{z} \leq \mathbf{y}\}$ and $V_T(\mathbf{y}) = (h \wedge \theta)(y_{n-1} + \cdots + y_1)$.

3. Optimality Analysis

In this section, we aim to characterize the structure of the optimal policy. Before that, we summarize and develop some important technical results regarding multimodularity that will be used in our subsequent optimality analysis.

3.1. Preliminaries: Multimodularity

Multimodularity is closely related to the concept of $L^\natural$-convexity. The two concepts can be transferred to each other through a linear transformation of independent variables (see Lemma 3 in Li and Yu 2014 for details). The relationship between the two concepts has also been reviewed in Pang et al. (2012). Li and Yu (2014) also develop some important properties of multimodularity. For example, Lemma 2 in Li and Yu (2014) states several preservation properties for multimodularity under certain common operators and Theorem 1 demonstrates that multimodularity can be preserved under a minimization operator, which plays a central role in optimality analysis for several lost-sales and perishable inventory systems studied by prior literature. One of the main preservation conditions is that the feasible region must be a multimodular polyhedron. This condition seems a bit restrictive for certain applications. For example, the feasible region $F_{\mathbf{d}}(\mathbf{x},\mathbf{D})$ (referring to Equation (1)) in our model may not be a multimodular polyhedron. The following two lemmas provide other preservation properties when the feasible region fails to be a multimodular polyhedron. The proofs of both lemmas and other results are relegated to Section A in Online Appendix.

LEMMA 1. *Assume that function $f(\mathbf{x}):\mathbb{R}_+^n \to \mathbb{R}$ is multimodular in $\mathbf{x}$ and satisfies the following condition: $\Delta_{x_n}^{\delta} f(\mathbf{x}) \leq \cdots \leq \Delta_{x_1}^{\delta} f(\mathbf{x})$. Then, the feasible region $F(\mathbf{x}) = \{\mathbf{z}|\mathbf{0} \leq \mathbf{z} \leq \mathbf{x}\}$ of the optimization problem*

$$g(\mathbf{x}) = \min_{\mathbf{z}\in F(\mathbf{x})}\{f(\mathbf{z})\} \tag{6}$$

can be replaced with $M(\mathbf{x}) = \{\mathbf{z}|\mathbf{z} \geq \mathbf{0}; \sum_{k=i}^{n} z_k \leq \sum_{k=i}^{n} x_k$ for $i \in [n].\}$ without effecting $g(\mathbf{x})$. In addition, $g(\mathbf{x})$ is multimodular in $\mathbf{x}$.

LEMMA 2. *Assume that $H(s):\mathbb{R}\to\mathbb{R}$ is a convex function and $f(\mathbf{x}):\mathbb{R}_+^n \to \mathbb{R}$ is a multimodular function, which satisfy the two conditions: a) $\Delta_{x_n}^{\delta} f(\mathbf{x}) \leq \cdots \leq \Delta_{x_m}^{\delta} f(\mathbf{x})$, for some $m \leq n$, and b) the function $H(\sum_{i=1}^{n} d_i) + f(\mathbf{x}-\mathbf{d})$ is decreasing in any d_i for any $i \in [n]$. Then, the feasible region $\Omega(\mathbf{x}, D, m)$ of the optimization problem*

$$g(\mathbf{x}) = \min_{\mathbf{d}\in\Omega(\mathbf{x},\, D,\, m)} \left\{ H\left(\sum_{i=1}^{n} d_i\right) + f(\mathbf{x}-\mathbf{d}) \right\} \tag{7}$$

can be replaced with $M(\boldsymbol{x}, D, m)$ *without effecting* $g(\boldsymbol{x})$, *where*

$$\begin{aligned} \Omega(\mathbf{x}, D, m) &= \left\{ \mathbf{d}\in\mathbb{R}^n \,\middle|\, \sum_{i=1}^{n} d_i \le D,\ 0 \le d_i \le x_i \text{ for } i\in[n], \text{ and } d_i = 0 \text{ for } i\in[m-1]. \right\}, \\ M(\mathbf{x}, D, m) &= \left\{ \mathbf{d}\in\mathbb{R}^n \,\middle|\, \sum_{j=1}^{i} d_j \le D \text{ for } i\in[n],\ d_i \le x_i \text{ for } i\in[n], \text{ and } d_i = 0 \text{ for } i\in[m-1] \right\}. \end{aligned} \tag{8}$$

In addition, $g(\mathbf{x})$ is multimodular in $\mathbf{x}$.

Remark 1. (Proof sketch of Lemmas 1 and 2). One can readily show that the feasible regions $\{(\mathbf{z}, \mathbf{x})|\mathbf{x}\in\mathbb{R}^n_+, \mathbf{z}\in F(\mathbf{x})\}$ of problem (6) may not be a multimodular polyhedron of $(\mathbf{z}, \mathbf{x})$. Therefore, Theorem 1 in Li and Yu (2014) cannot be directly applied. To address the problem, we first show that the feasible region $F(\mathbf{x})$ can be replaced by $M(\mathbf{x}) = \{\mathbf{z}|\mathbf{z}\ge\mathbf{0}; \sum_{k=i}^{n} z_k \le \sum_{k=i}^{n} x_k \text{ for } i\in[n]\}$ without changing the optimal solution of problem (6). However, the region $\{(\mathbf{z}, \mathbf{x})|\mathbf{x}\in\mathbb{R}^n_+, \mathbf{z}\in M(\mathbf{x}))\}$ is still not a multimodular polyhedron of $(\mathbf{z}, \mathbf{x})$. Hence, we further define new decision variables $\widehat{\mathbf{z}}$ as follows: $\widehat{z}_i = -z_i$, for any $i\in[n]$. The corresponding feasible region can be expressed as $\widehat{M}(\mathbf{x}) = \{\widehat{\mathbf{z}}|\widehat{\mathbf{z}}\le\mathbf{0}; \sum_{k=i}^{n} -\widehat{z}_k \le \sum_{k=i}^{n} x_k \text{ for } i\in[n]\}$, which is a multimodular polyhedron of $(\mathbf{x}, \widehat{z}_n, \ldots, \widehat{z}_1)$. As a result, it directly follows from Theorem 1 in Li and Yu (2014) that $g(\mathbf{x})$ is multimodular in $(x_1, \ldots, x_n)$. The proof of Lemma 2 has an analogous logic.

3.2. The Optimal Policy Structure

In this subsection, we will characterize the structure of the optimal policy. To this end, we make the following assumptions on system parameters.

Assumption 1. (i) $\theta + c \ge h$; (ii) $c \le p^w + h$.

Part (i) implies that it is cheaper to carry a unit to the next period than to dispose of it in the current period and purchase another unit in the next period. This assumption eliminates the trivial case in which it is always optimal to dispose of all on-hand inventories. Part (ii) means that the ordering cost is less than the sum of the shortage cost and the holding cost. In the lost-sales model, the shortage cost is typically greater than the ordering cost. Otherwise, if $c \ge p^w$, it will never be optimal to satisfy the demand.

To facilitate the subsequent analysis, we define $I_t^*(\mathbf{x}) = \min_{\mathbf{d}\in F_{\mathbf{d}}(\mathbf{x},\mathbf{D})} I_t(\mathbf{x}, \mathbf{d})$. The following pro position demonstrates that the objective functions $I_t^*(\mathbf{x})$, $u_t(\mathbf{x})$, $L_t(\mathbf{z})$ and $V_t(\mathbf{y})$ all have diagonal dominance and bounded sensitivity properties.

Proposition 1. (Properties of objective functions). *For each period t, the following inequalities hold:*

(i) $-p^1\delta \le \Delta^{\delta}_{x_n} I_t^* \le \Delta^{\delta}_{x_{n-1}} I_t^* \le \cdots \le \Delta^{\delta}_{x_1} I_t^*$;
(ii) $\Delta^{\delta}_{x_n} u_t - c\delta \le \Delta^{\delta}_{x_{n-1}} u_t \le \cdots \le \Delta^{\delta}_{x_1} u_t$;
(iii) $(h-\theta-c)\delta \le \Delta^{\delta}_{z_{n-1}} L_t \le \Delta^{\delta}_{z_{n-2}} L_t \le \cdots \le \Delta^{\delta}_{z_1} L_t$;
(iv) $(h-c)\delta \le \Delta^{\delta}_{y_{n-1}} V_t \le \Delta^{\delta}_{y_{n-2}} V_t \le \cdots \le \Delta^{\delta}_{y_1} V_t \le \theta\delta$.

These diagonal dominance properties essentially mean that a newer unit is more valuable than an older one. Similar results are also established by prior literature, such as Fries (1975), Nahmias (1975), Nansakumar and Morton (1993), and Li and Yu (2014). However, these papers did not consider multiple demand classes. In particular, Proposition 1 (iii) states that the value of inventory after disposal decreases with the inventory age, which implies that disposing of older inventories first is optimal, that is, the optimal disposal policy should follow the FIFO rule. Proposition 1 enables us to fully characterize the optimal fulfillment policy.

Theorem 1 (Optimal fulfillment policy). *The optimal fulfillment policy is characterized as follows.*

(i) ***Customer prioritization decision:*** $\Delta^{\delta}_{d_i^k} I_t\ pt < \Delta^{\delta}_{d_i^j} I_t$ *for* $i = (m^k \vee m^j), \ldots, n-1$ *if and only if* $p^k > p^j$, *which means that it is optimal to first satisfy the demand class with a higher shortage cost.*
(ii) ***Issuing decision:*** $\Delta^{\delta}_{d^k_{m^k}} I_t \le \Delta^{\delta}_{d^k_{m^k+1}} I_t \le \cdots \le \Delta^{\delta}_{d^k_n} I_t \le 0$ *for* $1 \le k \le w$, *which implies that it is optimal to first use older inventory to fulfill the demand, that is, the issuing policy follows the FIFO rule, and that each type of demand must be satisfied as much as possible.*

In brief, Theorem 1 says that customers of higher priority should be satisfied first and that the optimal inventory issuing policy for each demand class should follow the FIFO rule. The optimal demand fulfillment decision allows us to reformulate $u_t(\mathbf{x})$ in Equation (5) as a sequential decision process following the demand priority. Specifically, we define

$$u_t(\mathbf{z}, q) = cq + \alpha \mathbf{E}_{\mathbf{D}}\left\{\tilde{I}_t^1(\mathbf{z}, q)\right\}, \tag{9}$$

where the function $\tilde{I}_t^1(\mathbf{z}, q)$ can be recursively defined as follows: for $k = 1, \ldots, w$,

$$\begin{aligned} \tilde{I}_t^k(\mathbf{z}, q) &= \min_{\mathbf{d}^k \in \Omega((\mathbf{z}, q), D^k, m^k)} \left\{I_t^k(\mathbf{z}, q, \mathbf{d}^k)\right\}, \\ I_t^k(\mathbf{z}, q, \mathbf{d}^k) &= p^k\left(D^k - \sum_{i=1}^{n} d_i^k\right) + \tilde{I}_t^{k+1}((\mathbf{z}, q) - \mathbf{d}^k), \end{aligned} \tag{10}$$

where $\Omega(\mathbf{x}, D^k, m^k)$ is defined by Equation (8) and $\tilde{I}_t^{w+1}(\mathbf{z}, q) = \theta z_1 + V_{t+1}(z_2, \ldots, x_{n-1}, q)$.

COROLLARY 1. *The function $I_t^k(\mathbf{x})$ satisfies the following property: for any $k \in [w+1]$,*

$$\Delta^{\delta}_{x_{m^k}} I_t^k \le \Delta^{\delta}_{x_{m^{k+1}}} I_t^k \le \cdots \le \Delta^{\delta}_{x_n} I_t^k \le 0.$$

In what follows, we show that the objective functions in our model are multimodular, which essentially means that inventories of different ages are economic substitutes. With multimodularity, we can further derive the monotonicity and bounded sensitivity of the optimal disposal and replenishment decisions.

Define $q_t^*(\mathbf{y})$ as the optimal replenishment quantity and $\mathbf{z}_t^*(\mathbf{y})$ as the optimal inventory level after disposal. When there are multiple minimizers, we choose the smallest one in lexicographical order.

THEOREM 2 (Optimal disposal and replenishment policy).

(i) *The objective functions u_t and V_t are multimodular.*

(ii) *The optimal disposal policy can be characterized by $\bar{z}_{t,i}$, where $\bar{z}_{t,i}$ is a decreasing function of $y_{i+1}, y_{i+2}, \ldots, y_{n-2}$, and independent of $y_1, y_2, \ldots, y_i$. The optimal disposal policy is*

$$z_{t,\mathrm{i}}^*(\mathbf{y}) = \begin{cases} y_i, & \text{if } y_i \le \bar{z}_{t,i}; \\ \bar{z}_{t,i}, & \text{if } y_i > \bar{z}_{t,i}. \end{cases}$$

Moreover, the following inequalities hold:

$$-1 \le \Delta_{y_{i+1}} \bar{z}_{t,i} \le \Delta_{y_{i+2}} \bar{z}_{t,i} \le \cdots \le \Delta_{y_{n-2}} \bar{z}_{t,i} \le 0.$$

(iii) *The optimal replenishment quantity $q_t^*(\mathbf{y})$ is a decreasing function of $y_1, y_2, \ldots, y_{n-2}$, and the following inequalities hold:*

$$-1 \le \Delta_{y_{n-1}} q_t^* \le \Delta_{y_{n-2}} q_t^* \le \cdots \le \Delta_{y_1} q_t^* \le 0.$$

Theorem 2 (ii) shows that the optimal disposal policy follows the state-dependent base-stock structure, that is, it is optimal to dispose of the inventory with an i-period remaining lifetime down to a threshold level $\bar{z}_{t,i}$ if the inventory level is above $\bar{z}_{t,i}$; otherwise, it is optimal not to dispose. Moreover, the threshold level $\bar{z}_{t,i}$ depends only on newer inventories because a newer inventory can substitute for an older one in demand fulfillment, but not the reverse. The quantity $\bar{z}_{t,i}$ decreases in the on-hand inventory, whose remaining lifetime is longer than i periods and is more sensitive to the newer inventory. Analogously, the result (iii) shows that the order quantity and on-hand inventory of any age are economic substitutes and that the order quantity is more sensitive to the newer inventory. Although these results merely partially characterize the structure of the optimal policy, as seen later, they motivate us to design the efficient heuristic policy and relatively reduce the computational efforts.

4. Heuristic Policies

Although we have explored some properties of the objective functions and partially characterized the optimal policy structure, calculating the optimal policy still involves the curse of dimensionality. With the preceding optimality analysis, we intend to develop a nearly optimal and easy-to-calculate heuristic policy by approximating the optimal cost-to-go functions with a couple of single-variable functions. A similar idea has been successfully applied to other perishable inventory systems with a single demand class. For example, Li et al. (2016) use a linear function to approximate the profit-to-go function, while Sun et al. (2014) adopt a quadratic function. Our approach differs from theirs in two aspects. First, instead of using simple approximate functions, such as linear and quadratic functions, we construct adaptive approximate functions, which can be recursively calculated through a single-dimension dynamic program. Second, our approach not only ensures that the approximate function inherits the property of multimodularity, as the previous approaches did, but also requires the approximate function to satisfy diagonal dominance and bounded sensitivity properties characterized by Proposition 1.

Define $L_t^*(\mathbf{y}) = \min_{\mathbf{z} \in F_{\mathbf{z}}(\mathbf{y})}\{L_t(\mathbf{z})\}$. Referring to Equation (5), $V_t(\mathbf{y}) = \theta \sum_{i=1}^{n-1} y_i + L_t^*(\mathbf{y})$. Therefore, to approximate $V_t(\mathbf{y})$, it suffices to approximate $L_t^*(\mathbf{y})$. Following the earlier theoretical results, we require the approximate function $\widehat{L}_t^*(\mathbf{y})$ to satisfy the following two properties.

PROPERTY 1. (Theorem 2 (i)). *For any time period t, $\widehat{L}_t^*(\mathbf{y})$ is multimodular in $\boldsymbol{y}$.*

PROPERTY 2. (Proposition 1 (iii)). *For any time period t, $(h - c - \theta)\delta \le \Delta^{\delta}_{y_{n-1}} \widehat{L}_t^*(\mathbf{y}) \le \Delta^{\delta}_{y_{n-2}} \widehat{L}_t^*(\mathbf{y}) \le \cdots \le \Delta^{\delta}_{y_1} \widehat{L}_t^*(\mathbf{y}) \le 0$.*

Given the approximate function $\widehat{L}^*_{t+1}(\mathbf{y})$, we can derive the corresponding heuristic policy in period t by solving the following optimization problem.

$$\begin{aligned}
\bar{V}_t(\mathbf{y}) &= \theta \sum_{i=1}^{n-1} y_i + \bar{L}^*_t(\mathbf{y}) \\
\bar{L}^*_t(\mathbf{y}) &= \min_{\mathbf{z}\in F_{\mathbf{z}}(\mathbf{y}),\, q\in F_q} \left\{ (h-\theta)\sum_{i=1}^{n-1} z_i + cq + \alpha \mathbf{E}\left\{ \widehat{I}_t((\mathbf{z},\, q),\, \mathbf{d}^*((\mathbf{z},\, q),\, \mathbf{D})) \right\} \right\}, \\
\widehat{I}_t(\mathbf{x},\, \mathbf{d}) &= \theta\left(x_1 - \sum_{k=1}^{w} d_1^k \right) + \sum_{k=1}^{w} p^k \left(D^k - \sum_{i=1}^{n} d_i^k \right) + \widehat{V}_{t+1}\left(x_2 - \sum_{k=1}^{w} d_2^k,\, \ldots,\, x_n - \sum_{k=1}^{w} d_n^k \right), \\
\widehat{V}_{t+1}(\mathbf{y}) &= \theta \sum_{i-1}^{n-1} y_i + \widehat{L}^*_{t+1}(\mathbf{y}),
\end{aligned} \tag{11}$$

where $\mathbf{d}^*((\mathbf{z},\, q),\, \mathbf{D})$, characterized in Theorem 1, denotes the optimal demand fulfillment decision given disposal and ordering decisions $(\mathbf{z},\, q)$ and demand realizations $\mathbf{D}$).

The following Lemma 3 explains why our approach requires the approximate function to satisfy Properties 1 and 2. As such, the heuristic policy generated by problem (11) will inherit the same structure of the optimal policy characterized by Theorems 1 and 2.

Lemma 3. *The heuristic policy in every period t derived by problem* (11) *inherits the same structure as the optimal policy characterized in Theorems 1 and 2 if the approximate functions* $\widehat{L}^*_t(\mathbf{y})$ *satisfy Properties 1 and 2.*

Before introducing our approach in section 4.3, we first discuss how to adapt the two well-performing heuristics, that is, linear and quadratic approximations, for our problem, and they will serve as comparison benchmarks in numerical experiments.

4.1. Linear Approximation

Li et al. (2016) use a linear function $\widehat{L}^*_t(\mathbf{y}) = \sum_{i=1}^{n-1} a_i y_i$ to approximate the objective function $\bar{L}^*_t(\mathbf{y})$. Because a linear function is always a multimodular function, the approximation function $\widehat{L}^*_t(\mathbf{y})$ satisfies Property 1. In addition, $\widehat{L}^*_t(\mathbf{y})$ satisfies Property 2*if and only if* the coefficients $\{a_i\}_{i=1}^{n-1}$ satisfy $(h-c-\theta)\delta \le a_{n-1} \le \cdots \le a_1 \le 0$. Therefore, it is somewhat flexible for us to determine these coefficients. For instance, Li et al. (2016) set them all equal to the average of lower and upper bounds, that is, $a_1 = a_2 = \cdots = a_{n-1} = (h-c-\theta)/2$.

4.2. Quadratic Approximation

Sun et al. (2014) construct a $L^\natural$-convex quadratic function to approximate the value-to-go function. They show that for a backlog system, the construction of the approximate quadratic function is associated with a convex optimization and thus, can be easily solved. However, as we later show, such a construction for the lost-sale system is a non-convex optimization. Murota (2003) provides a specific form of quadratic functions that are $L^\natural$-convex. By the existing results (see Proposition 3 of Moriguchi and Murota (2019)), we can easily identify the quadratic functions that are multimodular.

Proposition 2 (Proposition 3 of Moriguchi and Murota 2019). *A quadratic function* $\mathcal{Q}(\mathbf{x};\mathbf{a},\, \mathbf{b},\, c) : \mathbb{R}^n \to \mathbb{R}$ *is multimodular in* $\mathbf{x}$ *if and only if it can be expressed as*

$$\mathcal{Q}(\mathbf{x};\mathbf{a},\, \mathbf{b},\, c) = \sum_{i=1}^{n}\sum_{j=1}^{i} Q^{ij}\left(\sum_{l=j}^{i} x_l; a^{ij},\, b^{ij} \right) + c, \tag{12}$$

where $Q^{ij}(s;a^{ij},b^{ij}) = a^{ij}\cdot s^2 + b^{ij}\cdot s$ *with* $a^{ij} \ge 0$, $b^{ij} \in \mathbb{R}$ *and* $c \in \mathbb{R}$.

Proposition 2 states that a multimodular quadratic function can be expressed as a summation of some quadratic functions (with summation of consecutive elements of $\mathbf{x}$ as variables). With the specific form of quadratic functions as basis, Sun et al. (2014) formulate the Bellman's equation of backlog systems as a linear program for infinite-horizon problems. However, their approach cannot be directly applied to our problem due to the following reasons. First, formulating the Bellman's equation as a linear programming is only applicable for infinite-horizon problems, where there is only one cost-to-go function to be approximated. In contrary, for finite-horizon problems, we need to approximate multiple objective functions which cannot be formulated as a linear programming. Second, a multimodular quadratic function does not satisfy Property 2 unless it is linear but unlike in the backlog systems, neither the single-period cost function nor the state dynamics is linear in decisions for lost-sales systems. For example, the fulfillment decision $\mathbf{d}^*((\mathbf{z},\, q),\, \mathbf{D})$ in Equation (11) may not be linear

Algorithm 1 Quadratic Approximation Approach.

Step 1. (State sampling) Sample states $\{\mathbf{y}_t^m\}_{m=1}^M$ for period $t=1,\cdots,T$ by simulating a reference policy π_r.

Step 2. (Initialization) Let $\widehat{L}_T^*(\mathbf{y})=(h\wedge\theta-\theta)\sum_{i=1}^{n-1}y_i$. t=T.

Step 3. (Generate state-cost pairs) Evaluate $\bar{L}_t^*(\mathbf{y}_t^m)$ at the sampled states $\{\mathbf{y}_t^m\}_{m=1}^M$ in period t by solving problem (11) to form state-cost pairs $\{(\mathbf{y}_t^m,\bar{L}_t^*(\mathbf{y}_t^m))\}_{m=1}^M$.

Step 4. (Estimation of coefficients $(\mathbf{a}_t,\mathbf{b}_t,c_t)$) Solve the following quadratic program to obtain the estimation of coefficients $(\mathbf{a}_t,\mathbf{b}_t,c_t)$ in period t.

$$\mathbf{a}_t,\mathbf{b}_t,c_t\in\arg\min_{\mathbf{a},\mathbf{b},c}\sum_{m=1}^{M}\left(\mathcal{Q}(\mathbf{y}_t^m;\mathbf{a},\mathbf{b},c)-\bar{L}_t^*(\mathbf{y}_t^m)\right)^2 \tag{13}$$

$$s.t.\ a^{ij}\geq 0,\forall i=1,\cdots,n-1,j=1,\cdots,i \tag{14}$$

$$h-c-\theta\leq\frac{\partial\mathcal{Q}(\mathbf{y}_t^m;\mathbf{a},\mathbf{b},c)}{\partial y_{n-1}}\leq\cdots\leq\frac{\partial\mathcal{Q}(\mathbf{y}_t^m;\mathbf{a},\mathbf{b},c)}{\partial y_1}\leq 0,\forall m=1,\cdots,M \tag{15}$$

Step 5. Set $\widehat{L}_t^*(\mathbf{y})=\mathcal{Q}(\mathbf{y}_t^m;\mathbf{a}_t,\mathbf{b}_t,c_c)$ to approximate $L_t^*(\mathbf{y})$. $t=t-1$. Return to Step 3 if $t>0$.

in ($\mathbf{z}$, q). Therefore, the resulting problem (11) could be non-convex. Following their ideas, we modify the quadratic approximation as follows.

Following Bertsekas (2011) and Sun et al. (2014), we solve the non-convex problem (11) with approximate dynamic programming, as reported in Algorithm 1. The state sampling in Step 1 follows the same idea as the counterpart in Sun et al. (2014), in which a set of states are generated by simulating a reference policy $\pi\pi_r$. Steps 2 to 5 construct the approximate quadratic functions inductively from period T to 1, which follow the same procedure as *approximation in value space* introduced by Bertsekas (2011). Construction of the quadratic function in each period is guided by the state-cost pairs $\{(\mathbf{y}_t^m,\bar{L}_t^*(\mathbf{y}_t^m))\}_{m=1}^M$ obtained by solving the approximated Bellman's equation (11).

In the quadratic programming (13)–(15), constraint (14) (also adopted by Sun et al. 2014) ensures the multimodularity of the approximate quadratic functions, while constraint (15) ensures that the diagonal dominance property holds at the sampled states $\{\mathbf{y}_t^m\}_{m=1}^M$. Notably, the diagonal dominance constraint (15) plays a similar role of regularization as for the supervised learning (Bertsekas 2011) to mitigate the problem of over-fitting. Indeed, we observed some evidences of over-fitting in the numerical experiments if constraint (15) is absent, especially when there exists multicollinearity among sampled states $\{\mathbf{y}_t^m\}_{m=1}^M$. Compared to the regularization suggested by Bertsekas (2011), where regularization parameters need to be tuned in every period, constraint (15) is clean and much easier to apply.

It is worthy noting that even with the approximate quadratic function obtained by imposing diagonal dominance constraint (15), the resulting problem (11) is still generally non-convex. Following Bertsekas (2011), we restrict the feasible region of decisions in problem (11) to evaluate the state-cost pairs in Step 3. Specifically, we confine the disposal decision to "never dispose" following Abouee Mehrizi et al. (2019), and apply the exhaustive search for the replenishment quantity. The exhaustive search makes the quadratic approximation more time consuming than our adaptive approximation.

4.3. Adaptive Approximation

Motivated by the quadratic approximation, we conduct an *adaptive approximation* approach by generalizing quadratic basis functions to more general forms. Under the adaptive approximation, the approximate function $\widehat{L}_{t+1}^*(\mathbf{y})$ is expressed as

$$\widehat{L}_{t+1}^*(\mathbf{y})=\sum_{i=1}^{n-1}\sum_{j=1}^{i}B_{t+1}^{ij}\left(\sum_{l=j}^{i}y_l\right), \tag{16}$$

where B_{t+1}^{ij} is referred to as *basis functions*.

We notice here that the basis functions $B_{t+1}^{ij}(\cdot)$ in our approach could be different over periods, and

Algorithm 2 Adaptive Approximation Approach.

Step 1. (Initialization) Let $B_{T,n-1}(s)=(h\wedge\theta-\theta)s$, $\eta_{T,1}=1$ and $\eta_{T,j}=0$ for $2\le j\le n-1$.

Step 2. (Recursion of Basis Functions) For any $1\le t\le T-1$, the basis functions $B_{t,n-1}(s)$ can be recursively calculated by solving the optimization problem (20) with $j=n-1$, where $\widehat{L}^*_{t+1}(\mathbf{y})=\sum_{j=1}^{n-1}\eta_{t+1,j}\cdot B_{t+1,n-1}(\sum_{l=j}^{n-1}y_l)$.

Step 3. (Determination of Weights $\eta_{t,j}$)

Step 3.1. Denote by D_t^s the total demand in period t, i.e., $D_t^s=\sum_{k=1}^{w}D_t^k$. Define U as the $1/w\sum_{k=1}^{w}(p^k-c)/(p^k+h)$ quantile of random total demand D_t^s. If $\Delta_s B_{t,n-1}(0)=1/U\cdot\Delta_s^U B_{t,n-1}(U)$, set $s'=U/2$; otherwise, find the smallest s' such that

$$\frac{1}{U}\int_0^U\left(B_{t,n-1}(s)-B_{t,n-1}(0)\right)/s\,ds=\left(B_{t,n-1}(s')-B_{t,n-1}(0)\right)/s'. \tag{23}$$

Step 3.2. For $j\in[n-1]$, let $\lambda_j=B_{t,j}(s')$ and determine weights $\eta_{t,j}$ as follows:

$$\eta_{t,j}=\frac{\lambda_j-\lambda_{j-1}}{B_{t,n-1}(s')-B_{t,n-1}(0)}, \tag{24}$$

where $\lambda_0=B_{t,n-1}(0)$.

Step 4. Let $\widehat{L}^*_t(\mathbf{y})=\sum_{j=1}^{n-1}\eta_{t,j}\cdot B_{t,n-1}(\sum_{l=j}^{n-1}y_l)$ to approximate $L^*_t(\mathbf{y})$ and turn to Step 2 for period $t-1$.

thus, the derived heuristic policy is non-stationary, while both Sun et al. (2014) and Li et al. (2016) consider mainly stationary cases.

When determining the basis functions, we face a tradeoff between heuristic efficiency and computation complexity. Including more basis functions would make the heuristic more efficient but may substantially increase the computation efforts. For example, in Equation (16), there are a total of $n(n-1)/2$ basis functions that may require a great deal of effort to calculate. We next discuss the idea behind the construction of basis functions in order to relatively reduce the computational complexity. Our approximation is essentially imposed on $\Delta_{y_i}L^*_{t+1}(\mathbf{y})$ rather than $L^*_{t+1}(\mathbf{y})$ itself.

• **The idea behind choosing the basis functions.** We first show that for any $j\in[n-1]$, the first partial derivative $\Delta_{y_j}L^*_{t+1}(\mathbf{y})$ can be bounded by

$$\begin{aligned}\Delta_{y_{n-1}}L^*_{t+1}\left(\left(\sum_{l=j}^{n-1}y_l\right)\mathbf{e}_{n-1}\right) &\le \Delta_{y_j}L^*_{t+1}\left(\left(\sum_{l=j}^{n-1}y_l\right)\mathbf{e}_{n-1}\right)\\ &\le \Delta_{y_j}L^*_{t+1}\left(\left(\sum_{l=1}^{j-1}y_l\right)\mathbf{e}_1+\left(\sum_{l=j}^{n-1}y_l\right)\mathbf{e}_{n-1}\right)\\ &\le \Delta_{y_j}L^*_{t+1}(\mathbf{y})\le 0,\end{aligned} \tag{17}$$

where the first and last inequalities follow from Property 2, the second from the convexity of $L^*_{t+1}(\mathbf{y})$, and the third from the multimodularity of $L^*_{t+1}(\mathbf{y})$.

Inequalities (17) show that $\Delta_{y_j}L^*_{t+1}(\mathbf{y})$ is bounded from below by $\Delta_{y_{n-1}}L^*_{t+1}\left(\left(\sum_{l=j}^{n-1}y_l\right)\mathbf{e}_{n-1}\right)$ and above by zero, which motivates us to use a linear combination of lower and upper bounds, that is, $\eta_{t+1,j}\cdot\Delta_{y_{n-1}}L^*_{t+1}\left(\left(\sum_{l=j}^{n-1}y_l\right)\mathbf{e}_{n-1}\right)$, to approximate $\Delta_{y_j}L^*_{t+1}(\mathbf{y})$. The way $\eta_{t+1,j}$ was selected will be discussed later. With this approximation of $\Delta_{y_j}L^*_{t+1}(\mathbf{y})$, we are further motivated to use $\eta_{t+1,j}\cdot L^*_{t+1}\left(\left(\sum_{l=j}^{n-1}y_l\right)\mathbf{e}_{n-1}\right)$ to approximate $L^*_{t+1}(\mathbf{y})$. That is, the function $L^*_{t+1}(\mathbf{y})$ can be approximated by

$$L^*_{t+1}(\mathbf{y})\approx\sum_{j=1}^{n-1}\eta_{t+1,j}\cdot L^*_{t+1}\left(\left(\sum_{l=j}^{n-1}y_l\right)\mathbf{e}_{n-1}\right)+c_{t+1}. \tag{18}$$

Because the constant simply longitudinally shifts the function and therefore does not affect the heuristic, we set c_{t+1}=0 in our approach. The function $\eta_{t+1,j}\cdot L^*_{t+1}\left(\left(\sum_{l=j}^{n-1}y_l\right)\mathbf{e}_{n-1}\right)$ can be viewed as the future

costs that correspond to the inventory with a lifetime no less than j periods.

We define $B_{t+1,j}(s)$ as an approximation of $L^*_{t+1}(s\mathbf{e}_j)$ for $j \in [n-1]$. By Equation (18), $L^*_{t+1}(\mathbf{y})$ can be re-approximated as

$$L^*_{t+1}(\mathbf{y}) \approx \widehat{L}^*_{t+1}(\mathbf{y}) = \sum_{j=1}^{n-1} \eta_{t+1,j} \cdot B_{t+1,n-1}\left(\sum_{l=j}^{n-1} y_l\right), \quad (19)$$

where $B_{t+1,j}(s)$ can be recursively calculated by the following dynamic program:

$$B_{t,j}(s) = \min_{\mathbf{z}\in F_{\mathbf{z}}(s\mathbf{e}_j),\, q\in F_q} \left\{(h-\theta)\sum_{i=1}^{n-1} z_i + cq + \alpha \mathbf{E}_{\mathbf{d}} \min_{\mathbf{D}\in F_{\mathbf{d}}((\mathbf{z},q),\,\mathbf{D})} \widehat{I}_t((\mathbf{z},q),\,\mathbf{d})\right\}, \quad (20)$$

where $\widehat{I}_t(\cdot)$ is defined by Equation (11) and $\widehat{L}^*_{t+1}(\mathbf{y})$ is given by Equation (19).

It is worth noting that the $n-1$ basis functions in Equation (19) share a common single-variable function $B_{t+1,n-1}(\cdot)$, which can be derived by solving the dynamic program (20) with $j = n-1$. Moreover, solving problem (20) is not difficult because it has a single dimension and inherits the same structure of the optimal policy (see Lemma 3). We next need to determine only the associated weights $\eta_{t+1,j}$ so that the approximation function $\widehat{L}^*_{t+1}(\mathbf{y})$ simultaneously satisfies Properties 1 and 2. Before discussing the idea behind the choice of weights, we present our adaptive approximation approach in Algorithm 2.

• **The idea of determining the weights.** We now discuss the idea behind the choice of weights in Equation (24). For $j \in [n-1]$, we can obtain by Equation (19) that $\Delta_{y_j} L^*_t(\mathbf{y}) \approx \sum_{l=1}^{j} \eta_{t,l} \cdot \Delta_s B_{t,n-1}\left(\sum_{i=l}^{n-1} y_i\right)$. It immediately follows that for $j \in [n-1]$,

$$\left(\Delta^{\delta}_{y_j} L^*_t(\mathbf{y}) - \Delta^{\delta}_{y_{j-1}} L^*_t(\mathbf{y})\right) \approx \eta_{t,j}\left(\Delta^{\delta}_s B_{t,n-1}\left(\sum_{i=j}^{n-1} y_i\right)\right), \quad (21)$$

where $\Delta^{\delta}_{y_0} L^*_t(\mathbf{y}) = 0$.

Setting $\mathbf{y} = \mathbf{0}$ in Equation (21), we have that for $j \in [n-1]$,

$$\left(L^*_t(\delta\mathbf{e}_j) - L^*_t(\delta\mathbf{e}_{j-1})\right) \approx \eta_{t,j}\left(B_{t,n-1}(\delta) - B_{t,n-1}(0)\right).$$

Replacing $L^*_{t+1}(s\mathbf{e}_j)$ with its approximate function $B_{t+1,j}(s)$ in the above equation, we can obtain

$$\left(B_{t,j}(\delta) - B_{t,j-1}(\delta)\right) \approx \eta_{t,j}\left(B_{t,n-1}(\delta) - B_{t,n-1}(0)\right). \quad (22)$$

Ideally, the weights $\eta_{t,j}$ should satisfy Equation (22) for any nonnegative δ. It is, however, not attainable, as $\eta_{t,j}$ in our approach are constants. To determine $\eta_{t,j}$, we specify $\delta = s'$ defined by Equation (23). Clearly, the constructed weights $\{\eta_{t,j}\}_{j=1}^{n-1}$ are a set of convex combinations.

The following lemma shows that the approximate functions $\widehat{L}^*_t(\mathbf{y})$ under the adaptive approximation approach satisfy Properties 1 and 2.

LEMMA 4. *The approximate functions $\widehat{L}^*_t(\mathbf{y})$ under the adaptive approximation approach satisfy Properties* 1 *and* 2, *and thus, the corresponding heuristic policy inherits the same structure characterized by Theorems* 1 *and* 2.

5. Numerical Experiment

In this section, we present a set of comprehensive numerical experiments. The goal of this numerical study is threefold. First, we test the effectiveness of the linear approximation proposed by Li et al. (2016), the quadratic approximation in Algorithm 1 adapted from Sun et al. (2014), the semi-myopic approximation proposed by Abouee Mehrizi et al. (2019) and our adaptive approximation. Second, we investigate the performance sensitivity of the heuristics with respect to some system primitives. Third, we apply the adaptive approximation to the LIFOM inventory system, that is, the case with multiple demand classes each with a least freshness requirement and following the LIFO issuing rule. Such a demand model is likely to arise in practice according to our customer behavior model in Online Appendix B.

5.1. Overall Performance Comparison

In our numerical studies, we consider two age-differentiated demand classes. Base system primitives are set according to the real data from the Sanquin blood bank in the Netherlands (see Table 2 in Haijema et al. 2007). To test more instances, we vary system primitives around the real values.

System Primitives. We fix the holding cost $h = 0.04$, the planning horizon $T = 7$ and the discount factor $\alpha = 0.99$ and test different values of several other primitives: the product lifetime n, the freshness requirements m^k, the shortage costs p^k, the disposal cost θ, and the ordering cost c. The complete set of these primitive values is given by $(n, m^1, m^2) \in \{(4, 1, 3), (5, 1, 4)\}$, $(p^1, p^2) \in \{(10, 30), (30, 30)\}$, $\theta \in \{0.05, 0.6, 3, 6\}$, and $c \in \{0.5, 1\}$. Using the real data from the Sanquin blood bank, Haijema et al. (2007) demonstrate that the demands across periods are typically non-stationary; there are peaks on Monday, Wednesday, and Friday. In our numerical studies, we also assume that demands of the two classes D^1_t and D^2_t in period t are independent but non-stationary across periods. The demands follow Poisson distributions with demand means described in Table 1. We consider two groups of demand distributions with high and low coefficients of variation, respectively. In each group, there are three types of demand stationarity: nonmonotonic, decreasing, and

Table 1 Demand Means in Each Period

	Demand stationarity	Demand means (D_t^1, D_t^2)
High C.V.	Nonmonotonic	{(2, 6), (2, 4), (2, 8), (2, 4), (2, 6), (2, 0), (2, 2)}
	Decreasing	{(2, 8), (2, 6), (2, 6), (2, 4), (2, 4), (2, 2), (2, 0)}
	Increasing	{(2, 0), (2, 2), (2, 4), (2, 4), (2, 6), (2, 6), (2, 8)}
Low C.V.	Nonmonotonic	{(8, 24), (8, 16), (8, 32), (8, 16), (8, 24), (8, 0), (8, 8)}
	Decreasing	{(8, 32), (8, 24), (8, 24), (8, 16), (8, 16), (8, 8), (8, 0)}
	Increasing	{(8, 0), (8, 8), (8, 16), (8, 16), (8, 24), (8, 24), (8, 32)}

increasing demands with respect to the time period. All combinations of described primitives give $2 \times 2 \times 4 \times 2 \times 6 = 192$ instances.

Performance Measure. To assess the performance of a specific heuristic, we compare its total costs with the optimal costs. The initial system state is set to zero, that is, $\mathbf{y}_1 = 0$. In particular, we define the relative performance gap of heuristic policy P as follows:

$$\text{Performance(P)} = \frac{V_1^{\text{P}} - V_1}{V_1} \times 100\%,$$

where V_1 represents the optimal cost and V_1^{P} denotes the total expected costs under the heuristic policy P, with P being adaptive approximation, linear approximation, quadratic approximation, or semi-myopic approximation.

Table 2 summarizes the overall performance of the adaptive, linear, quadratic, and semi-myopic approximations. It can be seen that adaptive and quadratic approximation are both nearly optimal and significantly outperform the other two approximation approaches. In particular, over all the testing instances, the average optimality gap of the adaptive (quadratic) approximation is only 0.30% (0.42%) with the worst optimality gap 2.16% (9.01%). The semi-myopic approximation with the average optimality gap 6.14% performs better than the linear approximation, which has an average optimality gap 16.08%. Moreover, the linear approximation performs poorly in the worse case, with a maximal gap of 72.03%. The "std" means the standard deviation of the optimality gap and reflects the robustness of the heuristic policy. The stds of adaptive, quadratic, and semi-myopic approximations are only 0.4%, 0.91%, and 1.47%, respectively, implying that all of them perform quite robustly over the testing instances. In contrast, the linear approximation with the std of 16.18% is less satisfactory. Overall, the adaptive approximation approach demonstrates consistently superior performance. Quadratic approximation also works well, albeit less robust as indicated by the maximal optimality gap (9.01%); we will examine this aspect further in the next subsection.

Table 2 Overall Performance (%) of Adaptive, Linear, and Semi-Myopic Heuristics

Heuristic	Max	Mean	Min	Std.
Adaptive approx.	2.16	0.30	0.00	0.40
Linear approx.	72.03	16.08	1.44	16.18
Semi-myopic approx.	9.45	6.14	0.25	1.47
Quadratic approx.	9.01	0.42	0.01	0.91

5.2. Sensitivity of Model Primitives

In this subsection, we investigate the effect of system primitives on the performance of the four heuristics. Table 3 illustrates the heuristic effectiveness with respect to the coefficient of variation, types of demand stationarity and shortage costs. We observe that all heuristics but the semi-myopic approximation perform less effectively under a higher demand uncertainty; nevertheless, the relative performance comparison among the heuristics remains the same irrespective of the level of demand uncertainty. It can also be observed that the effectiveness of all heuristics seems robust to types of demand stationarity. Moreover, as the shortage cost increases, the adaptive approximation performs slightly worse, while the other three heuristics perform better. Nevertheless, all heuristics appear robust to the shortage cost, and the level of shortage cost does not change the relative performance among the heuristics either.

Table 4 reports the performance of the four heuristics with respect to disposal cost and expiration cost. We report two different cases. In the first case, as in the theoretical analysis, we assume the unit disposal cost and unit expiration cost are equal, corresponding to columns 3 to 6. In the second case, we assume the unit disposal cost and unit expiration cost to be unequal, denoted (θ_d, θ_e), corresponding to columns 7 to 10 in the table; this case is particularly tested to investigate the robustness of quadratic approximation, as mentioned previously in Section 5.1. We find that the effectiveness of adaptive approximation is quite robust to the variation of disposal cost and expiration cost, in both cases, although there is a slight increase in optimality gap when the cost parameters increase. In contrast, the performance of the other heuristics is very sensitive to the variation of unit

Table 3 Performance (%) of Different Heuristics Under Different Primitives

	Adaptive approx.				Linear approx.				Semi-myopic approx.				Quadratic approx.			
Group (Num. of inst.)	**Max**	**Mean**	**Min**	**Std.**	**Max**	**Mean**	**Min**	**Std.**	**Max**	**Mean**	**Min**	**Std.**	**Max**	**Mean**	**Min**	**Std.**
High c.v.(96)	2.16	0.51	0.05	0.48	72.03	17.90	1.44	18.05	7.12	5.53	3.98	0.89	2.36	0.49	0.12	0.34
Low c.v.(96)	0.37	0.08	0.00	0.08	50.78	14.25	1.45	13.91	9.45	6.74	2.53	1.68	9.01	0.35	0.01	1.25
Nonmonotonic(64)	1.67	0.34	0.04	0.35	54.46	14.29	1.45	14.46	7.64	5.39	2.53	0.99	1.23	0.23	0.03	0.25
Decreasing(64)	1.65	0.21	0.01	0.31	72.03	17.14	1.44	17.32	9.37	6.89	4.19	1.35	9.01	0.54	0.02	1.15
Increasing(64)	2.16	0.34	0.00	0.51	70.62	16.80	1.58	16.71	9.45	6.12	3.79	1.63	6.30	0.50	0.01	1.04
p^1=10(96)	1.79	0.24	0.00	0.36	72.03	16.75	1.44	16.96	9.45	6.69	4.26	1.33	9.01	0.59	0.01	1.25
p^1=30(96)	2.16	0.35	0.00	0.44	58.52	15.41	1.45	15.42	9.22	5.58	2.53	1.40	1.09	0.25	0.01	0.26

Table 4 Performance (%) of Adaptive, Linear, and Semi-Myopic Heuristics Under Different θ

θ (θ_d, θ_e)		**0.05**	**0.6**	**3**	**6**	**(0.05, 6.05)**	**(0.6, 6.6)**	**(3, 9)**	**(6, 12)**
Number of instances		**48**	**48**	**48**	**48**	**24**	**24**	**24**	**24**
Adaptive approx.	Max	0.70	0.59	2.16	1.82	0.34	0.58	0.78	1.30
	Mean	0.12	0.14	0.35	0.59	0.08	0.12	0.32	0.35
	Min	0.00	0.00	0.02	0.02	0.00	0.00	0.06	0.07
	Std.	0.14	0.14	0.41	0.55	0.09	0.13	0.23	0.30
Linear approx.	Max	4.19	9.01	36.01	72.03	3.61	7.36	29.41	58.21
	Mean	2.17	4.81	19.36	37.97	2.00	4.35	17.24	34.20
	Min	1.44	2.75	10.55	21.18	1.27	2.47	9.32	19.22
	Std.	0.60	1.20	6.66	13.76	0.55	1.03	5.24	11.52
Semi-myopic approx.	Max	9.45	9.25	8.61	9.37	33.26	27.86	21.35	17.85
	Mean	6.44	6.21	5.91	5.97	12.28	11.04	9.31	8.51
	Min	4.02	3.37	2.53	3.79	4.39	4.38	4.37	4.30
	Std.	1.65	1.52	1.26	1.43	8.51	6.90	5.12	4.28
Quadratic approx.	Max	0.52	2.46	3.35	9.01	36.26	27.60	11.99	22.41
	Mean	0.19	0.28	0.35	0.86	9.20	6.76	3.23	3.85
	Min	0.01	0.01	0.01	0.01	0.02	0.01	0.01	0.01
	Std.	0.16	0.42	0.53	1.62	11.66	8.49	3.96	5.25

Table 5 The Performance of Linear and Adaptive Approximation with Single Demand Class (%)

	(1, 5)		(1, 10)		(1, 1.3)		(1.2, 1.3)	
(c, r) = lifetime	**3**	**4**	**3**	**4**	**3**	**4**	**3**	**4**
c.v.=0.2								
θ=0.2	0.08(0.09)	0.11(0.10)	0.01(0.01)	0.01(0.01)	0.58(0.00)	0.87(0.00)	2.07(0.01)	2.25(0.00)
θ=0.5	0.05(0.05)	0.11(0.10)	0.01(0.01)	0.01(0.01)	0.55(0.00)	3.1(0.00)	2.06(0.01)	11.28(0.01)
θ=0.9	0.06(0.05)	0.15(0.14)	0.01(0.01)	0.01(0.01)	0.46(0.00)	2.99(0.00)	2.06(0.02)	11.25(0.02)
c.v.=0.3								
θ=0.2	0.04(0.03)	0.05(0.03)	0.03(0.03)	0.11(0.11)	1.02(0.00)	1.48(0.00)	3.73(0.02)	4.05(0.00)
θ=0.5	0.05(0.03)	0.04(0.02)	0.03(0.03)	0.15(0.15)	0.98(0.00)	3.1(0.00)	3.72(0.02)	11.28(0.01)
θ=0.9	0.05(0.03)	0.05(0.03)	0.07(0.07)	0.11(0.11)	0.96(0.00)	2.99(0.00)	3.7(0.02)	11.25(0.02)
c.v.=0.4								
θ=0.2	0.08(0.06)	0.08(0.05)	0.10(0.09)	0.17(0.16)	1.61(0.00)	2.24(0.00)	6.26(0.01)	6.78(0.00)
θ=0.5	0.02(0.00)	0.02(0.00)	0.09(0.08)	0.15(0.14)	1.53(0.00)	0.86(0.00)	6.22(0.02)	2.25(0.00)
θ=0.9	0.04(0.01)	0.04(0.01)	0.06(0.05)	0.13(0.12)	1.43(0.00)	2.99(0.00)	6.19(0.02)	11.25(0.02)
c.v.=0.5								
θ=0.2	0.18(0.15)	0.18(0.14)	0.14(0.13)	0.17(0.16)	2.21(0.00)	3.16(0.00)	10.38(0.03)	11.29(0.01)
θ=0.5	0.20(0.17)	0.2(0.16)	0.11(0.10)	0.15(0.14)	2.09(0.00)	0.86(0.00)	10.27(0.05)	2.25(0.00)
θ=0.9	0.23(0.20)	0.23(0.19)	0.12(0.11)	0.21(0.20)	1.94(0.00)	2.99(0.00)	10.13(0.09)	11.25(0.02)

disposal and expiration costs. As the cost parameters increase, the effectiveness of linear approximation quickly deteriorates. Semi-myopic and quadratic approximations perform better when the unit disposal cost and expiration cost are equal, but deteriorate significantly when the two cost parameters are unequal, in terms of both absolute performance (the optimality gap is much bigger in the second case

Table 6 The Performance of Linear and Adaptive Approximation with Multiple Demand Classes (%)

	(1, 5)		(1, 10)		(1, 1.3)		(1.2, 1.3)	
(*c*, *r*) lifetime	3	4	3	4	3	4	3	4
c.v.=0.2								
θ=0.2	0.01(0.00)	0.01(0.00)	0.00(0.00)	0.00(0.00)	0.11(0.02)	0.68(0.10)	2.07(0.10)	2.24(0.39)
θ=0.5	0.01(0.00)	0.01(0.00)	0.00(0.00)	0.00(0.00)	0.59(0.02)	0.76(0.11)	2.06(0.10)	2.24(0.39)
θ=0.9	0.01(0.00)	0.01(0.00)	0.00(0.00)	0.00(0.00)	0.54(0.03)	0.74(0.10)	2.05(0.10)	2.24(0.38)
c.v.=0.3								
θ=0.2	0.02(0.00)	0.02(0.00)	0.00(0.00)	0.00(0.00)	0.16(0.00)	1.05(0.05)	3.75(0.11)	3.95(0.51)
θ=0.5	0.02(0.00)	0.02(0.00)	0.00(0.00)	0.00(0.00)	1.03(0.06)	1.12(0.07)	3.73(0.13)	3.93(0.51)
θ=0.9	0.02(0.00)	0.02(0.00)	0.00(0.00)	0.00(0.00)	0.84(0.02)	1.11(0.07)	3.71(0.17)	3.92(0.51)
c.v.=0.4								
θ=0.2	0.03(0.00)	0.02(0.00)	0.01(0.00)	0.00(0.00)	0.36(0.00)	1.55(0.04)	6.21(0.04)	6.31(0.34)
θ=0.5	0.03(0.00)	0.02(0.00)	0.01(0.00)	0.00(0.00)	1.50(0.04)	1.59(0.05)	6.15(0.07)	6.27(0.32)
θ=0.9	0.03(0.00)	0.02(0.00)	0.01(0.00)	0.00(0.00)	1.26(0.02)	1.58(0.05)	6.09(0.14)	6.24(0.31)
c.v.=0.5								
θ=0.2	0.03(0.00)	0.03(0.00)	0.01(0.00)	0.01(0.00)	0.95(0.08)	2.19(0.11)	9.87(0.06)	10.00(0.18)
θ=0.5	0.03(0.00)	0.03(0.00)	0.01(0.00)	0.01(0.00)	1.86(0.05)	2.16(0.10)	9.68(0.07)	9.92(0.15)
θ=0.9	0.03(0.00)	0.03(0.00)	0.01(0.00)	0.01(0.00)	1.89(0.01)	2.16(0.09)	9.48(0.09)	9.85(0.15)

when the two cost parameters are unequal than in the first case) and sensitivity to the parameter variation.

5.3. Adaptive Approximation for LIFO Systems

In the previous analysis, our model assumes that the firm can control the demand fulfillment and thus, the optimal issuing policy follows the FIFO rule. In some other situations such as in offline retailing, FIFO may not always be the proper choice model given that consumers control the item depletion process. In the literature, it is widely assumed that in offline retailing, fresher items are sold first in a LIFO manner. Li et al. (2016) are the first to study the optimal policy structure of perishable inventory systems with the LIFO issuing rule in the setting of single demand class. In Online Appendix C, we extend the optimality analysis and apply our adaptive approximation approach to the case of multiple demand classes with the LIFO issuing rule. In this subsection, we numerically examine the performance of adaptive approximation approach and conduct the comparison with the linear approximation proposed by Li et al. (2016), For comparison convenience, we adopt their profit-maximizing model, instead of the cost-minimizing model in our FIFO system; however, the two models are essentially equivalent.

Consistent with Li et al. (2016), we fix the horizon length $T = 10$, the discount factor $\alpha = 0.95$, the holding cost $h = 0.2$, and the clearance price $s = 0.4$. Demands follow the uniform distribution with the mean equal to 5. We test various values of other parameters as follows: the product lifetime $n \in \{3, 4\}$, the demand coefficients of variation c.v. $\in \{0.2, 0.3, 0.4, 0.5\}$, the ordering cost and the selling price $(c, r) \in \{(1, 5), (1, 10), (1, 1.3), (1.2, 1.3)\}$, and the outdating cost $\theta \in \{0.2, 0.5, 0.9\}$. In the setting of multiple demand classes, we consider two demand classes, D^1 and D^2, with the least freshness requirement m^1=1 and $m^2 = n-1$, respectively. Demands of the two classes are equal with a total size following the uniform distribution with mean 5.

Tables 5 and 6 summarize numerical results for single and multiple demand classes, respectively. The numbers in and out of the parenthesis represent the relative error of adaptive and linear approximation heuristics, respectively. It can be seen that our adaptive approximation heuristic performs consistently well with the worst performance less than 0.51% and performs much better than the linear approximation heuristic. There are three observations regarding the performance sensitivity to system parameters. First, the linear approximation also performs quite well for instances with large gross margin $r - c$, but not for instances with low gross margin. However, our adaptive approximation is not sensitive to the gross margin. Second, as stated in Li et al. (2016), the performance of linear approximation deteriorates as the product lifetime becomes longer. Our numerical results demonstrate that the adaptive approximation seems insensitive to the product lifetime. Third, both heuristics perform better for instances with low demand variability, while the adaptive approximation is more robust to demand variability.

6. Conclusion

In this study, we have studied a perishable inventory system with age-differentiated demand, which has wide applications, including online and offline B2C settings, B2B settings, and blood inventory management. By establishing new properties of multimodularity, we

derive the structural results for the optimal policies and show that the optimal issuing policy follows the FIFO rule. The optimality analysis motivates us to propose the adaptive approximation heuristic, which can be recursively calculated through a single-dimension dynamic program. Numerical studies demonstrate that our proposed approximation approach is nearly optimal with the average optimality gap 0.30% and significantly outperforms the existing heuristics studied in the literature. Our optimality analysis and heuristic idea can also be applied to the case with the LIFO issuing rule.

Our model assumes that the selling price is exogenously determined. One of the important future research directions is to incorporate the pricing decision into the perishable inventory system with multiple demand classes. Customers are sensitive to both the selling price and the product freshness. It would be interesting to analyze periodic promotion/pricing decisions across age groups of products. The problem will become quite challenging; however, the technique tool developed by Chen et al. (2014) may be helpful. Our model implicitly assumes that customers simultaneously arrive at offline or online stores. However, in practice, customers typically arrive at stores sequentially. A natural direction is to model customers' arrival as a sequential arrival process. Our results can be extended to the case of sequential arrival when the shortage costs of different demand classes are identical (e.g., see Online Appendix C for handling the LIFOM system). This assumption is not restrictive in retailing because the shortage cost originates mainly from the revenue loss, which is the same for all customers. However, taking pricing into account will make the problem completely different and quite challenging. Designing efficient and easy-to-implement heuristic policies would prove valuable both practically and theoretically.

Acknowledgments

The authors thank the Department Editor Annabelle Feng, the Associate Editor, and two anonymous reviewers for their guidance and constructive comments. Y. Li acknowledges the support of Hong Kong Research Grant Council (GRF 9042254) and City University of Hong Kong (Project 9360147). Y. Yang thanks the financial support from the National Natural Science Foundation of China [Grants 71871198 and 71931009]. Y. Yang and W. Zhou are grateful for the financial support from the National Natural Science Foundation of China [Grants 71571160 and 71821002]. All authors have an equal contribution to the study.

Notes

[1]https://www.amazon.com/Organic-Valley-Pasteurized-Gallon-Ounces/product-reviews/B000O6EG9G/ref=cm_cr_dp_d_hist_1?ie=UTF8&filterByStar=one_star&reviewerType=all_reviews-reviews-filter-bar

[2](https://item.jd.com/3979213.html#none

[3]For online retailing, the retailer may choose to treat all customers the same as they pay the same price, or, because it can distinguish different customers, the retailer may choose to assign higher penalty costs to customers of higher frequency, bigger baskets, or high end. This, however, makes no difference to our modeling and solution.

References

Abouee Mehrizi, H., O. Baron, O. Berman, D. Chen. 2019. Managing perishable inventory systems with multiple priority classes. *Prod. Oper. Manag.* **28**(9): 2305–2322.

Akkaş, A. 2019. Shelf space selection to control product expiration. *Prod. Oper. Manag.* **28**(9): 2184–2201.

Bertsekas, D. P. 2011. *Dynamic Programming and Optimal Control*, 3rd edn, Vol. **II**. Athena Scientific, Belmont, MA.

Buck, R., A. Minvielle. 2013. A Fresh Take on Food Retailing. McKinsey report. Available at https://www.mckinsey.com/~/media/mckinsey/dotcom/client_service/retail/articles/perspectives\%20-%20winter%202013/3_fresh_take_on_food_retailing_vf.ashx (accessed date March 1, 2018).

Buzby, J., H. Farah-Wells, J. Hyman. 2014. The estimated amount, value, and calories of postharvest food losses at the retail and consumer levels in the United States. *Economic Information Bulletin 164262, United States Department of Agriculture, Economic Research Service*.

Chao, X., X. Gong, C. Shi, H. Zhang. 2015. Approximation algorithms for perishable inventory systems. *Oper. Res.* **63**(3): 585–601.

Chao, X., X. Gong, C. Shi, C. Yang, H. Zhang, S. X. Zhou. 2017. Approximation algorithms for capacitated perishable inventory systems with positive lead times. *Management Sci.* **64**(11): 5038–5061.

Chen, X., Z. Pang, L. Pan. 2014. Coordinating inventory control and pricing strategies for perishable products. *Oper. Res.* **62**(2): 284–300.

Chen, S., Y. Li, W. Zhou. 2019. Joint decisions for blood collection and platelet inventory control. *Prod. Oper. Manag.* **28**(7): 1674–1691.

Civelek, I., I. Karaesmen, A. Scheller-Wolf. 2015. Blood platelet inventory management with protection levels. *Eur. J. Oper. Res.* **243**(3): 826–838.

Cohen, M. A., D. Pekelman. 1978. LIFO inventory systems. *Management Sci.* **24**(11): 1150–1162.

Deniz, B., I. Karaesmen, A. Scheller-Wolf. 2010. Managing perishables with substitution: Inventory issuance and replenishment heuristics. *Manuf. Serv. Oper. Manag.* **12**(2): 319–329.

Food and Agriculture Organization (FAO). 2016. Key Facts on Food Loss and Waste You Should Know. Available at the website of Food and Agriculture Organization of the United Nations, http://www.fao.org/save-food/resources/keyfindings/en/ (accessed date May 3, 2017).

Fries, B. E. 1975. Optimal ordering policy for a perishable commodity with fixed lifetime. *Oper. Res.* **23**(1): 46–61.

Fu, K., X. Gong, G. Liang. 2019. Managing perishable inventory systems with product returns and remanufacturing. *Prod. Oper. Manag.* **28**(6): 1366–1386.

Goh, C.-H., B. S. Greenberg, H. Matsuo. 1993. Two-stage perishable inventory models. *Management Sci.* **39**(5): 633–649.

Ha, A. Y. 1997. Inventory rationing in a make-to-stock production system with several demand classes and lost sales. *Management Sci.* **43**(8): 1093–1103.

Haijema, R., J. van der Wal, N. M. van Dijk. 2005. Blood platelet production: A multi-type perishable inventory problem. H.

Fleuren, D. den Hertog, P. Kort, eds. *Operations Research Proceedings 2004*. Springer, Berlin, Heidelberg, 84–92.

Haijema, R., J. van der Wal, N. M. van Dijk. 2007. Blood platelet production: Optimization by dynamic programming and simulation. *Comput. Oper. Res.* **34**(3): 760–779.

Ishii, H., T. Nose. 1996. Perishable inventory control with two types of customers and different selling prices under the warehouse capacity constraint. *Int. J. Prod. Econ.* **44**(1): 167–176.

Karaesmen, I. Z., A. Scheller-Wolf, B. Deniz. 2011. Managing perishable and aging inventories: Review and future research directions. K. Kempf, P. Keskinocak, R. Uzsoy, eds. *Planning Production and Inventories in the Extended Enterprise*. Springer, Berlin, 393–436.

Lebersorger, S., F. Schneider. 2014. Food loss rates at the food retail, influencing factors and reasons as a basis for waste prevention measures. *Waste Manag.* **34**(11): 1911–1919.

Li, Q., P. Yu. 2014. Multimodularity and its applications in three stochastic dynamic inventory problems. *Manuf. Serv. Oper. Manag.* **16**(3): 455–463.

Li, Y., A. Lim, B. Rodrigues. 2009. Pricing and inventory control for a perishable product. *Manuf. Serv. Oper. Manag.* **11**(3): 538–542.

Li, Q., P. Yu, X. Wu. 2016. Managing perishable inventories in retailing: Replenishment, clearance sales, and segregation. *Oper. Res.* **64**(6): 1270–1284.

Li, Q., P. Yu, X. Wu. 2017. Shelf life extending packaging, inventory control and grocery retailing. *Prod. Oper. Manag.* **26**(7): 1369–1382.

Moriguchi, S., K. Murota. 2019. On fundamental operations for multimodular functions. *J. Oper. Res. Soc. Japan* **62**(2): 53–63.

Murota, K. 2003. *Discrete Convex Analysis*. SIAM, Philadelphia.

Nahmias, S. 1975. Optimal ordering policies for perishable inventory—II. *Oper. Res.* **23**(4): 735–749.

Nahmias, S. 1976. Myopic approximations for the perishable inventory problem. *Management Sci.* **22**(9): 1002–1008.

Nahmias, S. 1977. Higher-order approximations for the perishable-inventory problem. *Oper. Res.* **25**(4): 630–640.

Nahmias, S. 1982. Perishable inventory theory: A review. *Oper. Res.* **30**(4): 680–708.

Nansakumar, P., T. E. Morton. 1993. Near myopic heuristics for the fixed-life perishable problem. *Management Sci.* **39**(12): 1490–1498.

Pang, Z., F. Y. Chen, Y. Feng. 2012. A note on the structure of joint inventory-pricing control with leadtimes. *Oper. Res.* **60**(3): 581–587.

Pierskalla, W. P., C. D. Roach. 1972. Optimal issuing policies for perishable inventory. *Management Sci.* **18**(11): 603–614.

Sun, P., K. Wang, P. Zipkin. 2014. Quadratic Approximation of Cost Functions in Lost Sales and Perishable Inventory Control Problems. Technical report, Working Paper. Fuqua School of Business, Duke University, Durham, NC.

Zhang, H., C. Shi, X. Chao. 2016. Approximation algorithms for perishable inventory systems with setup costs. *Oper. Res.* **64**(2): 432–440.

Zhang, H., J. Zhang, R. Q. Zhang. 2020. Simple policies with provable bounds for managing perishable inventory. *Prod. Oper. Manag.* **29**(11): 2637–2650.

Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Appendix A. Proofs for the Main Results.

Appendix B. Customer Behavior Models for Perishable Inventory.

Appendix C. Multiclass LIFO Inventory System and Empirical Validation of Multiclass FIFO assumption.